ACCESS FORM

Design Document

Version 1.0

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Revision History

Date (dd/mm/yyyy)	Version	Description	Author
12/12/2025	1.0	Access Form is an easy and accessible form builder and survey system made for all users, especially disabled and elderly people. The system has three main parts: Admin Panel for managing users and surveys, Form Creator Panel for creating and publishing surveys, and Respondent Panel for filling surveys. The system follows WCAG 2.1 accessibility rules, so users can use keyboard navigation, screen readers, high contrast mode, and change font size easily. All screens are designed in a simple way so that users do not face difficulty while using the system. The design document includes system architecture, interface designs, use cases, sequence diagrams, and test cases. The main purpose of Access Form is to make surveys easy and accessible for everyone.	BC220203921

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1 Introduction of Design Document

This Design Document describes the design of **Access Form**, which is an accessibility focused form builder and survey system. The system is mainly designed for **disabled persons**, such as visually impaired users, physically disabled users, elderly people, and users who find complex systems difficult to use.

The main purpose of Access Form is to provide a **simple, easy, and accessible platform** where surveys can be created and filled without difficulty. The system focuses on reducing user effort and making sure that everyone, including disabled users, can use the system comfortably.

1.1 System Users

The system is divided into three main user panels:

- **Admin Panel**
  The Admin Panel is used to manage the overall system. The admin can manage users, monitor surveys, generate reports, and make sure the system works properly for all users.
- **Form Creator Panel**
  The Form Creator Panel allows users to register first and then log in to create surveys. Form creators can add questions, apply accessibility options such as font size and contrast, publish surveys, and view responses.
- **Respondent Panel**
  The Respondent Panel is designed for users who fill surveys. This panel is especially helpful for disabled persons because it supports simple navigation, keyboard usage, and screen reader support.

1.2 Accessibility Focus

Access Form is designed to support users with different needs. The system provides features like adjustable font size, high contrast display, keyboard navigation, and simple screen layout. These features help disabled and elderly users to understand and complete surveys easily.

1.3 Purpose of This Document

This Design Document explains how the system is designed. It includes diagrams, database design, interface design, and test cases. Each part of the design shows how accessibility and user roles (Admin, Form Creator, and Respondent) are handled in the system.

2 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) represents the complete data structure of the Access Form system. The **User** entity is the core entity of the system, which stores information of all users. Each user can have a specific role such as **Admin, Form Creator, or Respondent**. The **Form Creator** is responsible for creating and managing surveys. Each survey consists of multiple questions, which are stored in the **Question** entity. The **Respondent** participates in surveys and submits responses. Each response is stored in the **Response** entity and contains multiple answers linked to individual questions through the **Answer** entity. To support disabled users, the system includes an **Accessibility Settings** entity, which stores accessibility related preferences such as font size, high contrast mode, screen reader support, and keyboard navigation. This ERD ensures proper data organization and supports accessibility requirements of the system.

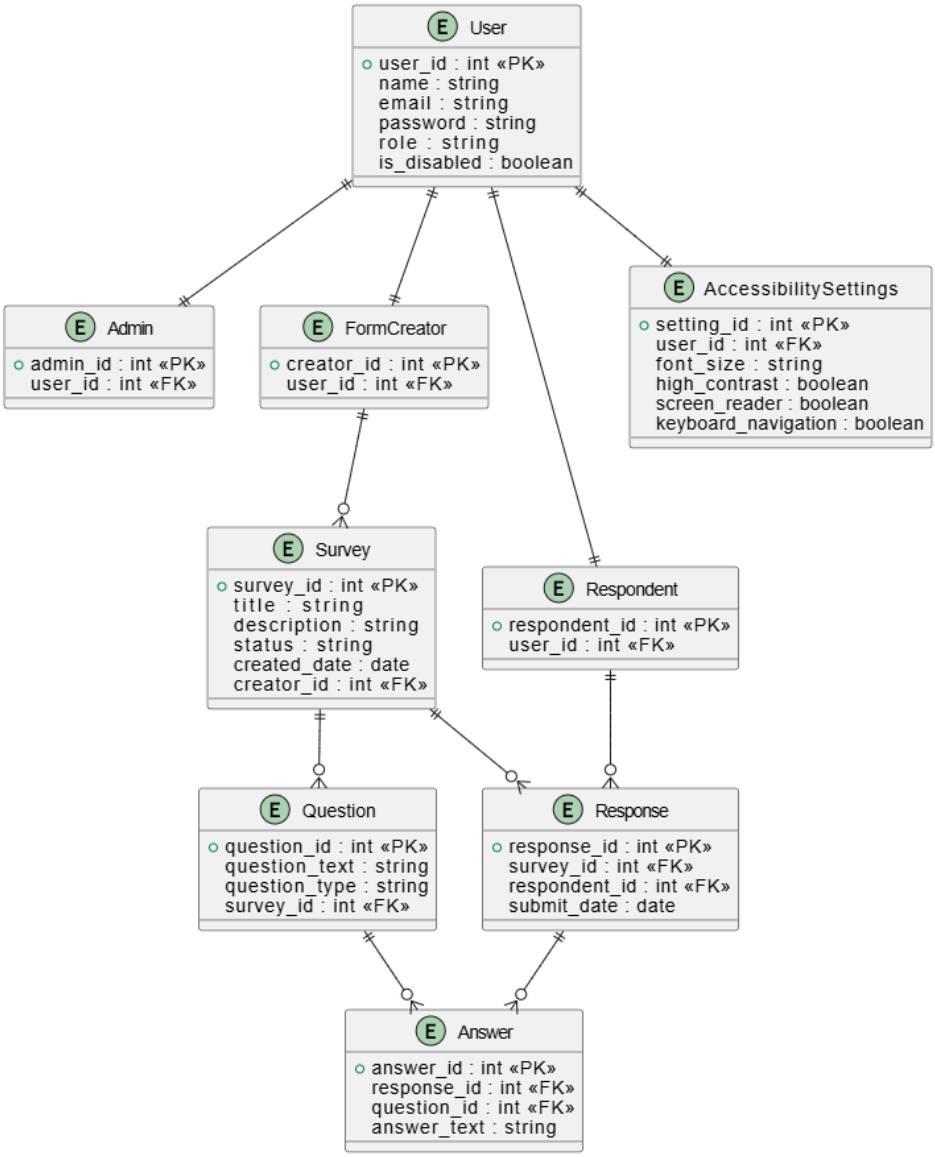


Figure 1 - Entity Relationship Diagram (ERD)

3 Sequence Diagrams

3.1 User Registration

This sequence diagram explains how a new user registers in the Access Form system. First, the user opens the registration screen and enters basic details such as name, email, password, and selects the user role (Admin, Form Creator, or Respondent). After clicking the register button, the system sends the entered data to the user management process for validation. The system checks whether all required fields are filled correctly and whether the email already exists in the database. If the information is valid and the email is not registered before, the system saves the new user data in the database and shows a successful registration message. If the email already exists or required information is missing, the system displays an error message and asks the user to correct the details. This process ensures secure and proper user registration before login.

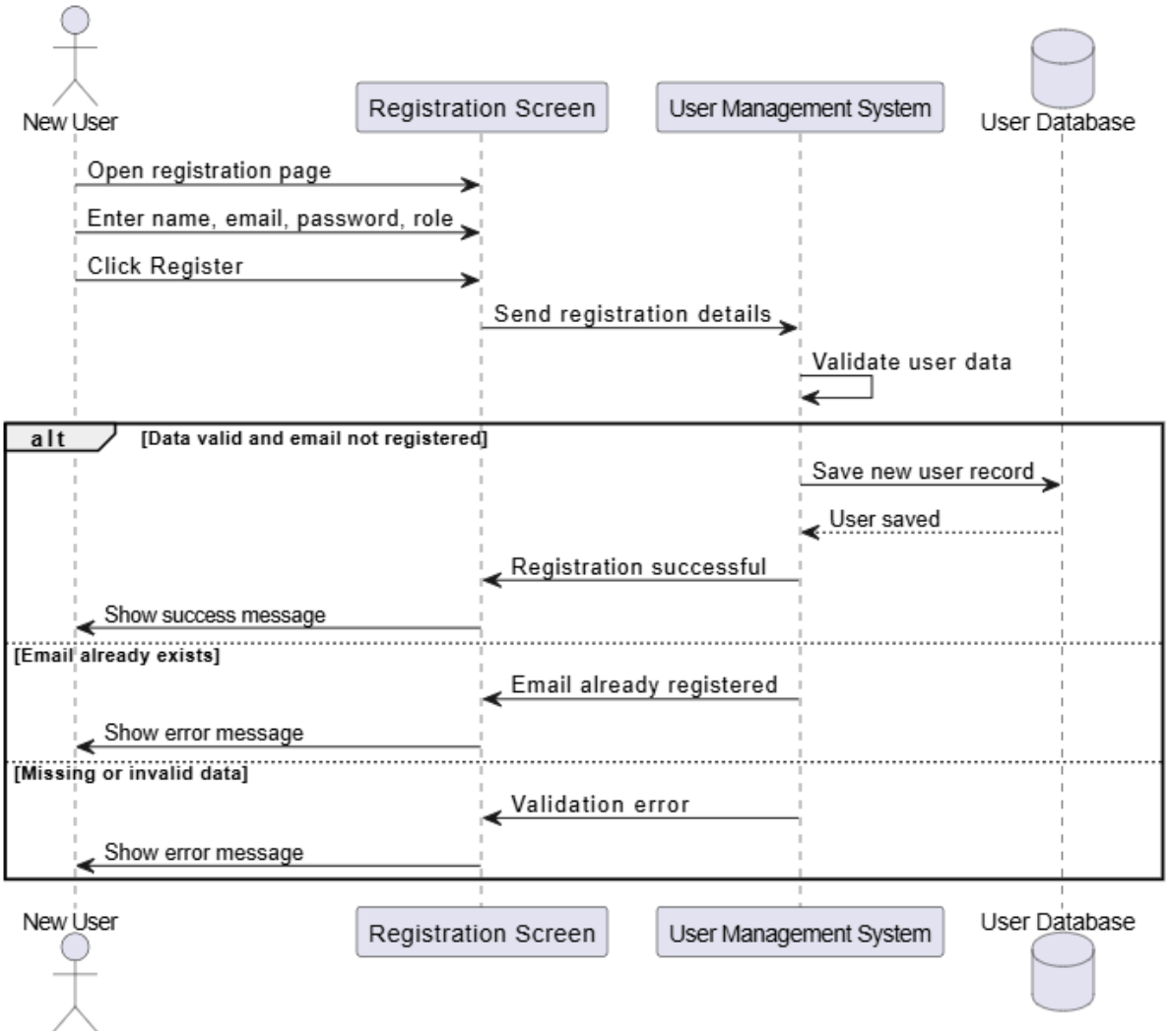


Figure 2 - Sequence Diagram for User Registration

3.2 User Login

This sequence diagram explains how a registered user logs into the Access Form system. First, the user opens the login screen and enters the registered email and password. After clicking the login button, the system sends the entered credentials to the user management process for verification. The system checks the email and password with the data stored in the database. If the credentials are correct, the system allows the user to log in and redirects the user to the respective panel such as Admin Panel, Form Creator Panel, or Respondent Panel. If the email or password is incorrect, the system shows an error message and asks the user to enter valid login details. This process ensures secure access to the system.

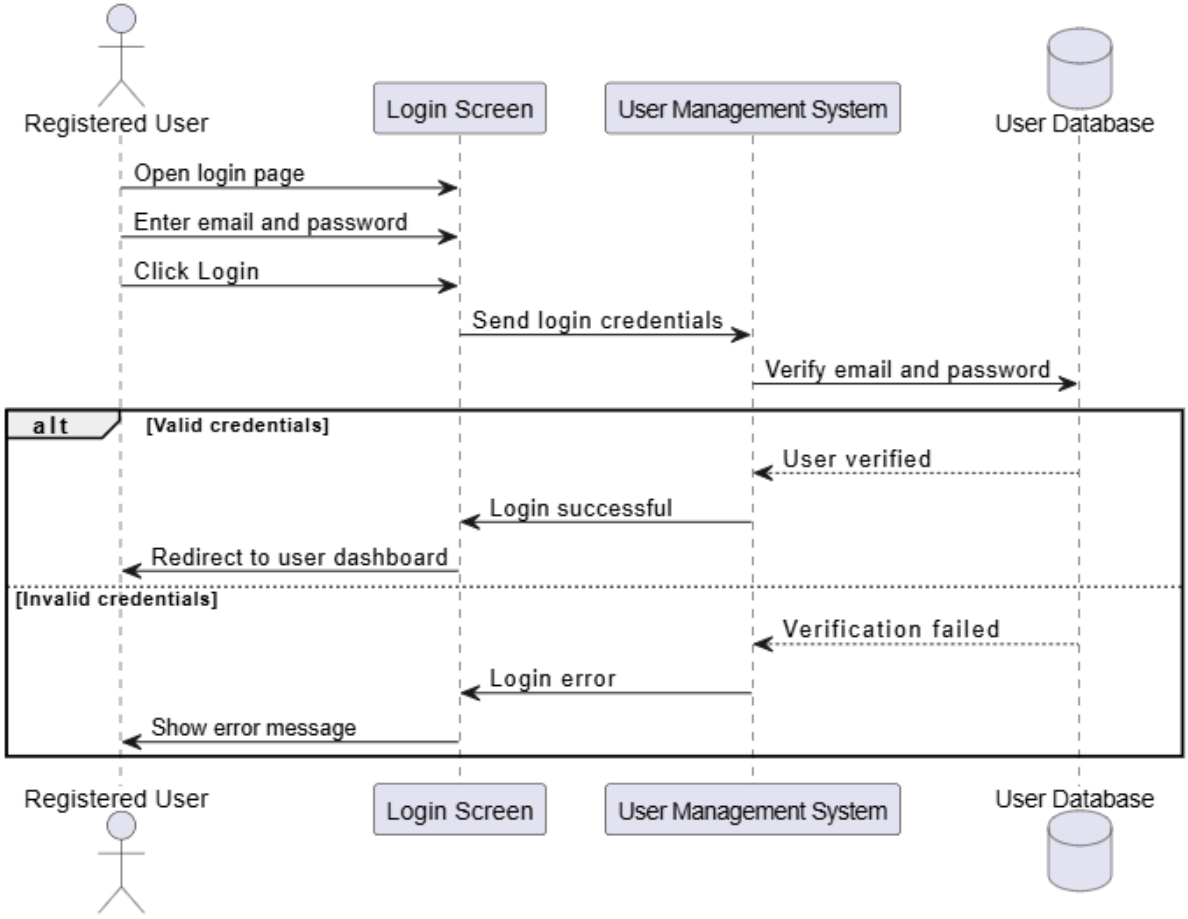


Figure 3 - Sequence Diagram for User Login

3.3 Manage Accessibility Settings

This sequence diagram explains how a user manages accessibility settings in the Access Form system. First, the user opens the Accessibility Settings option from the system interface. The system displays available options such as large text size, high contrast mode, keyboard navigation, and screen reader support. The user selects or updates the required options and clicks the save button. The system validates the selected settings and stores them in the user preferences database. After saving, the system applies the settings immediately and shows a confirmation message. If any problem occurs while saving the settings, the system shows an error message and asks the user to try again. This process helps disabled users use the system more easily.

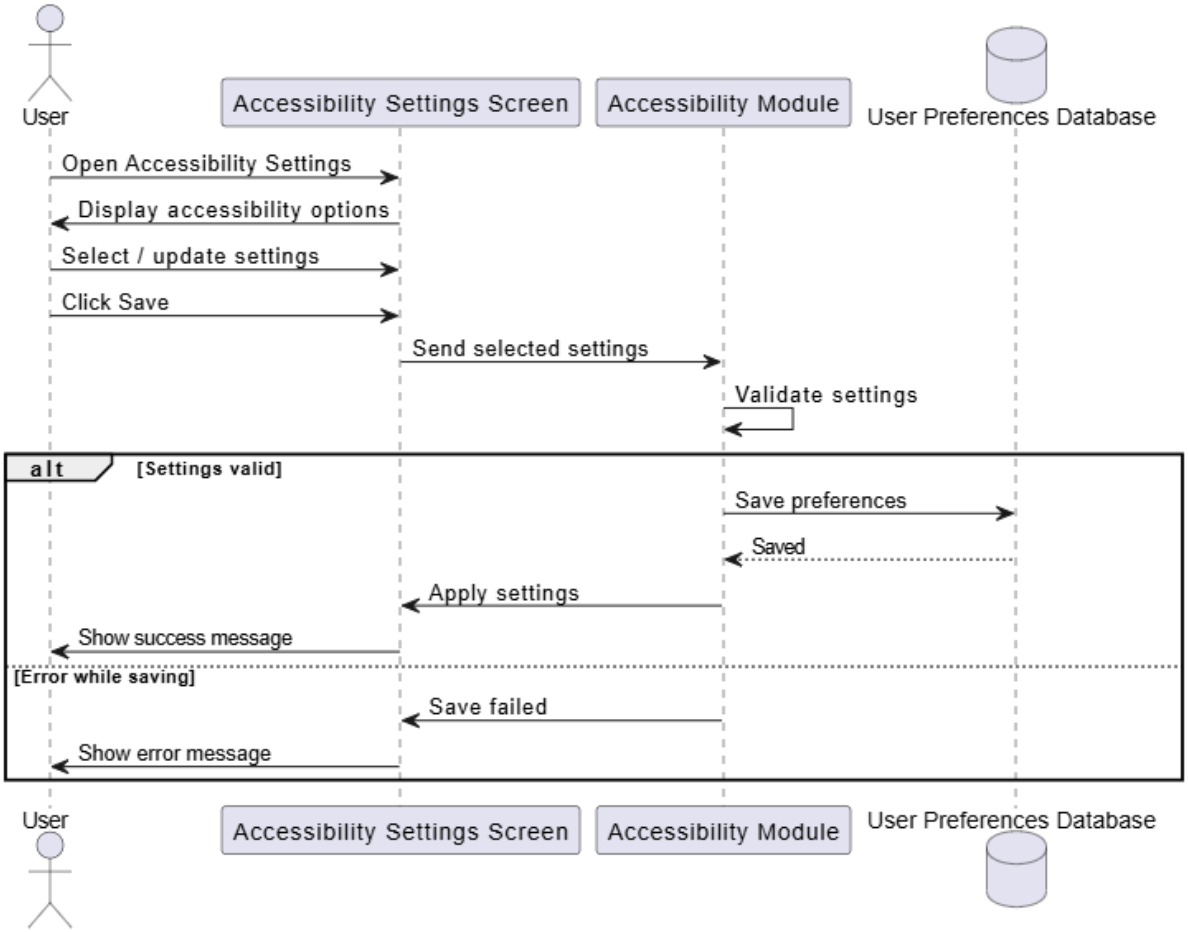


Figure 4 - Sequence Diagram to Manage Accessibility Settings

3.4 ***Manage Users (Admin Panel)***

This sequence diagram explains how the admin manages users in the Access Form system. First, the admin opens the Manage Users option from the Admin Panel. The system displays the list of registered users with their current status. The admin selects a user and performs actions such as view details, activate, or deactivate the user account. The system sends the selected action to the user management process. The system updates the user status in the database and shows a confirmation message to the admin. If any error occurs during updating the user status, the system shows an error message. This process helps the admin control and manage system users properly.

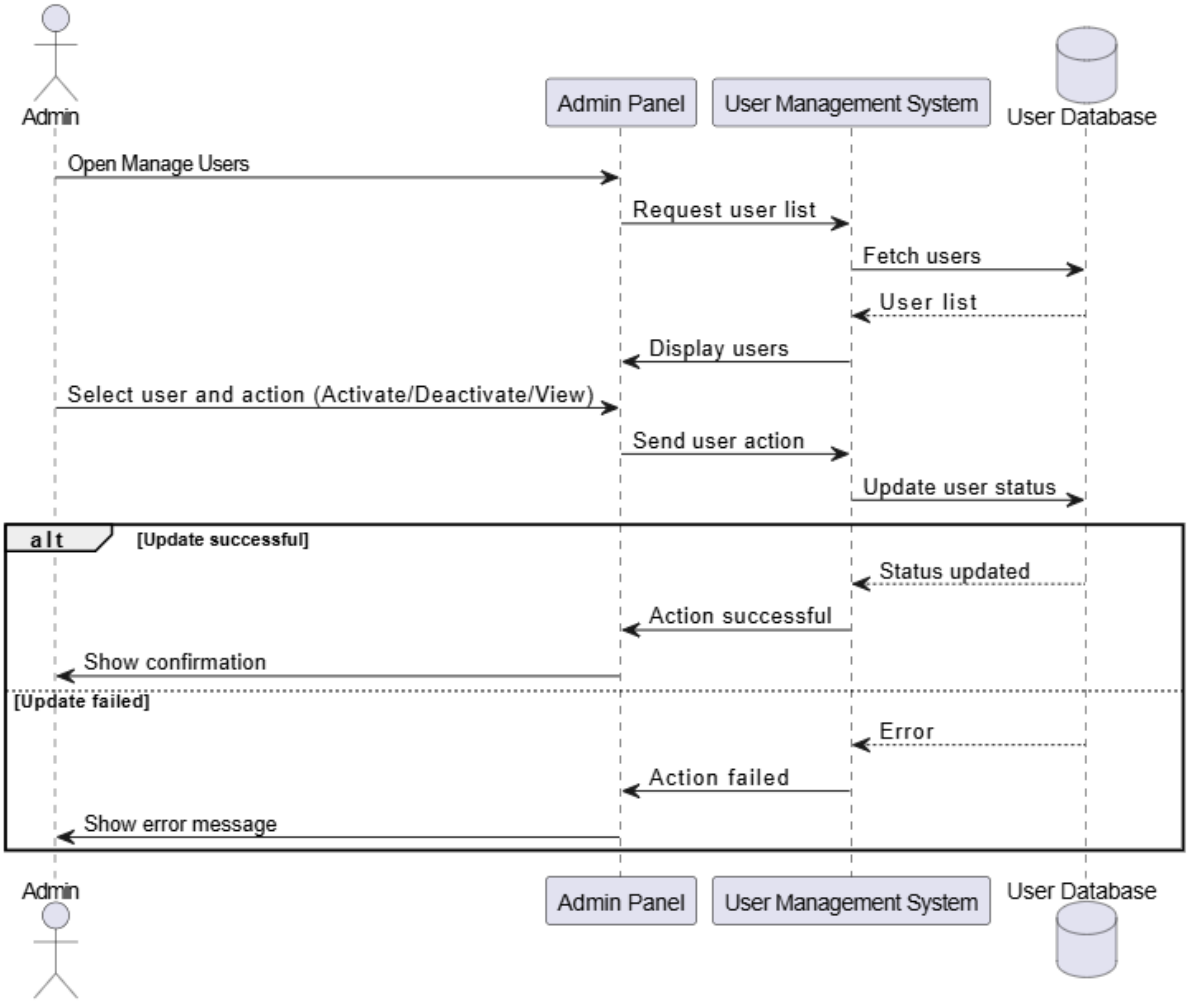


Figure 5 - Sequence Diagram to Manage Users

3.5 ***View All Survey (Admin Panel)***

This sequence diagram explains how the admin views all surveys in the Access Form system. First, the admin opens the “View All Surveys” option from the Admin Panel. The system sends a request to the survey management process to fetch all available surveys. The survey management process retrieves the survey list from the database and sends it back to the system interface. The system displays the list of surveys to the admin with basic details such as survey title, creator name, and status. If the survey list is successfully loaded, the admin can view the details of any survey. If there is any problem while fetching the surveys, the system shows an error message to the admin. This process helps the admin monitor all surveys created in the system.

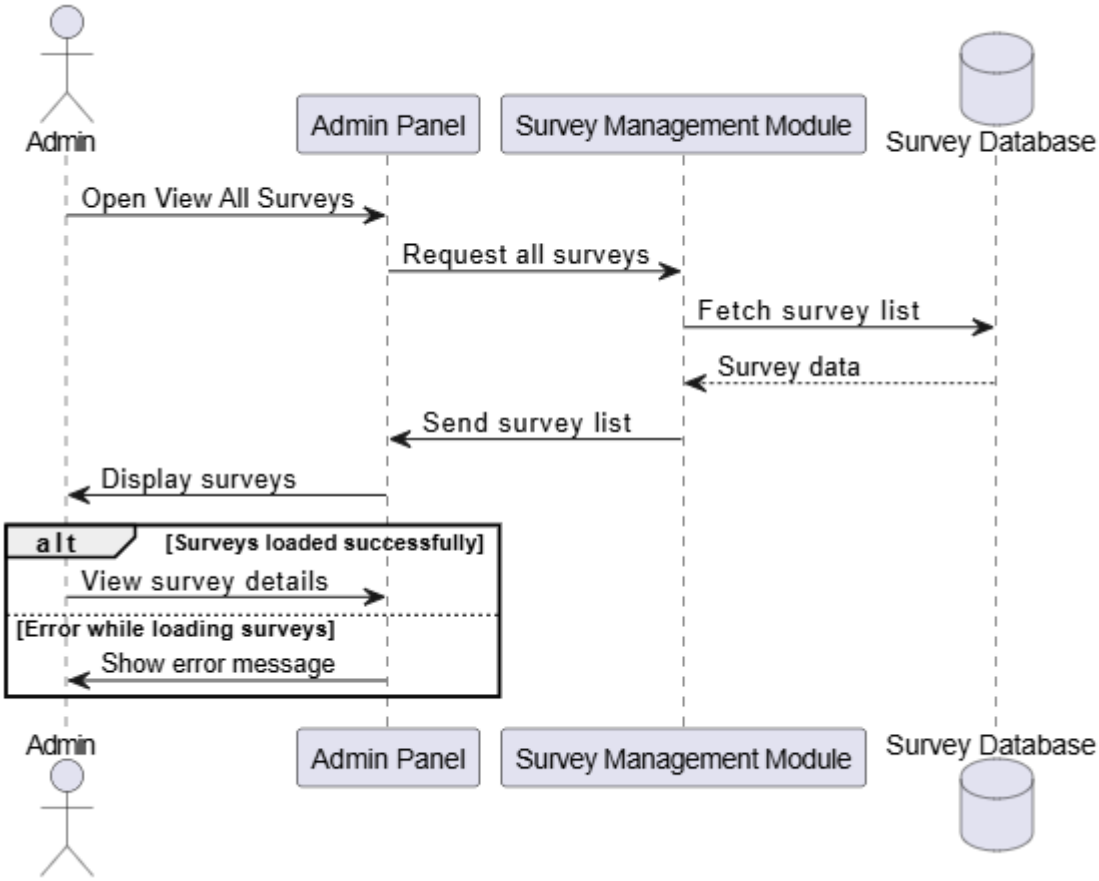


Figure 6 - Sequence Diagram to view All Surveys

3.6 View Survey Responses / Reports (Admin Panel)

This sequence diagram explains how the admin views survey responses and reports in the Access Form system. First, the admin opens the “View Survey Responses / Reports” option from the Admin Panel. The system asks the admin to select a survey. After selecting a survey, the system sends a request to the survey management process to fetch all responses related to that survey. The survey management process retrieves the responses from the database and sends them back to the system interface. The system displays the responses in a readable format such as list or summary report. If the responses are loaded successfully, the admin can review them for analysis. If there is any issue while fetching the responses, the system shows an error message. This process helps the admin analyze survey results effectively.

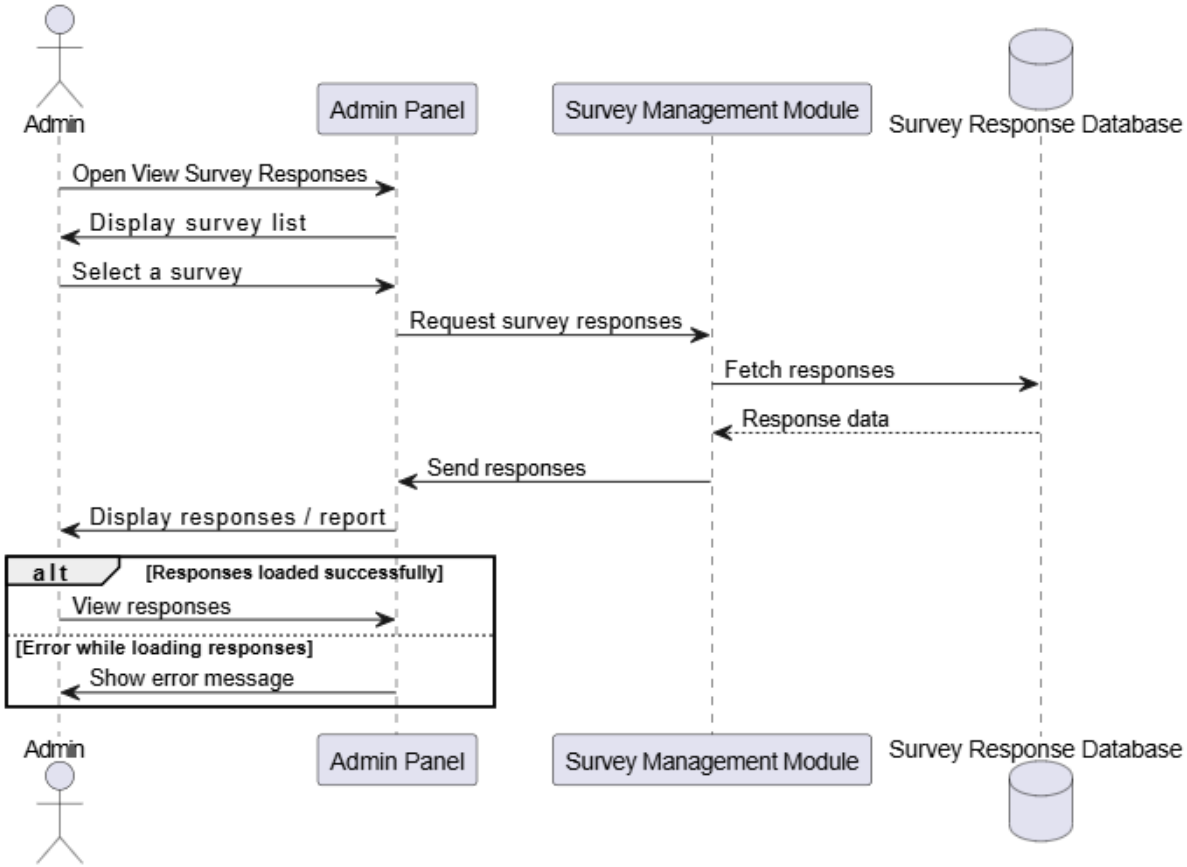


Figure 7 - Sequence Diagrams to View Survey Responses / Reports

3.7 Survey Creation (Form Creator Panel)

This sequence diagram explains how a form creator creates a new survey in the Access Form system. First, the form creator opens the “Create New Survey” option from the Form Creator Panel. The system displays the survey creation form where the form creator enters the survey title and description. After entering the basic information, the form creator adds questions to the survey. Once the survey details and questions are added, the form creator chooses to save the survey as a draft or publish it. The system validates the survey information and saves the survey data in the database. If the survey is saved or published successfully, the system shows a confirmation message. If any required information is missing or an error occurs, the system shows an error message and asks the form creator to correct the details. This process allows form creators to easily create surveys.

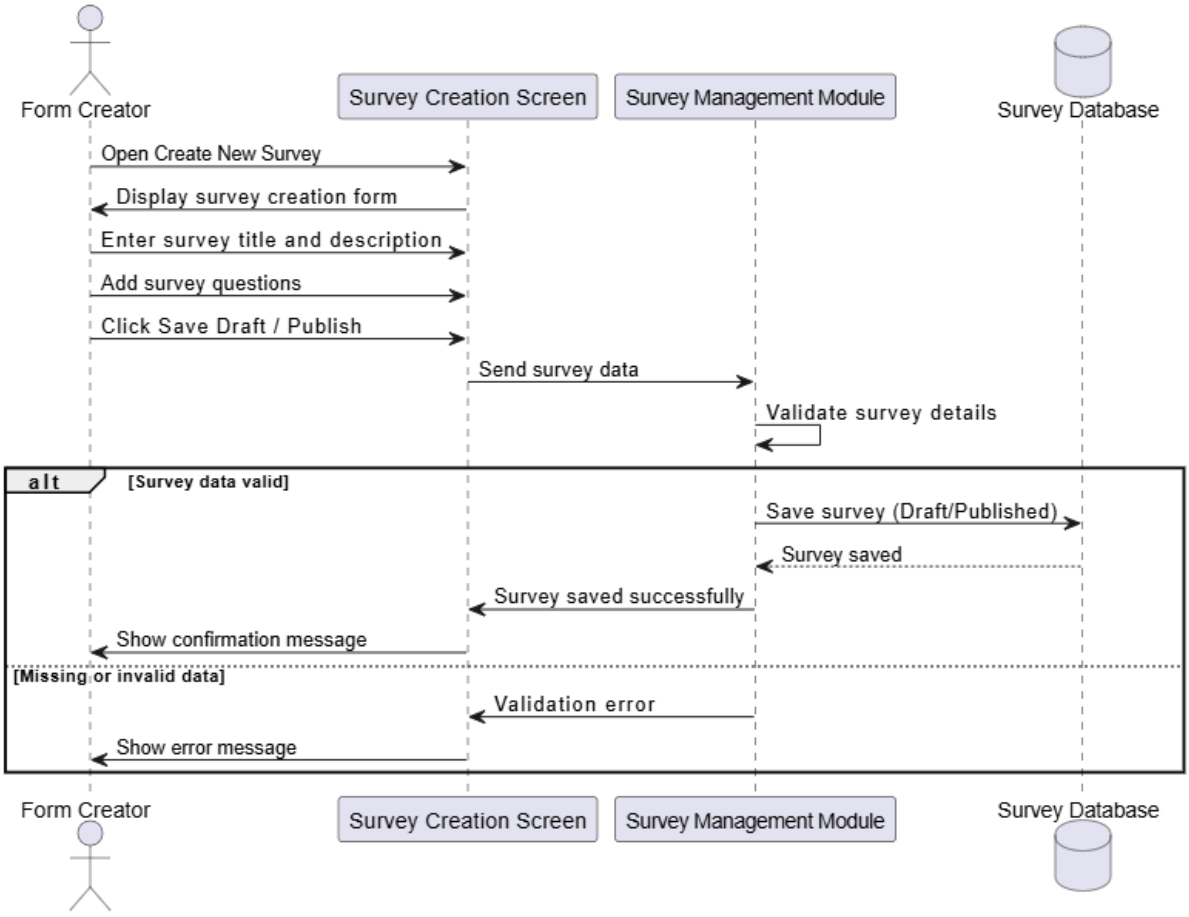


Figure 8 - Sequence Diagram for Survey Creation

3.8 Add Questions (Form Creator Panel)

This sequence diagram explains how a form creator adds a new question to a survey in the Access Form system. First, the form creator opens an existing survey or a new survey in the survey creation screen. The system displays the current survey details. The form creator selects the option to add a new question and enters the question text, question type (such as multiple choice or text), and options if required. After entering the details, the form creator saves the question. The system validates the question data and stores it in the database linked with the survey. If the question is added successfully, the updated survey is shown on the screen. If any required information is missing or an error occurs, the system shows an error message and asks the form creator to correct the details. This process allows form creators to build surveys step by step.

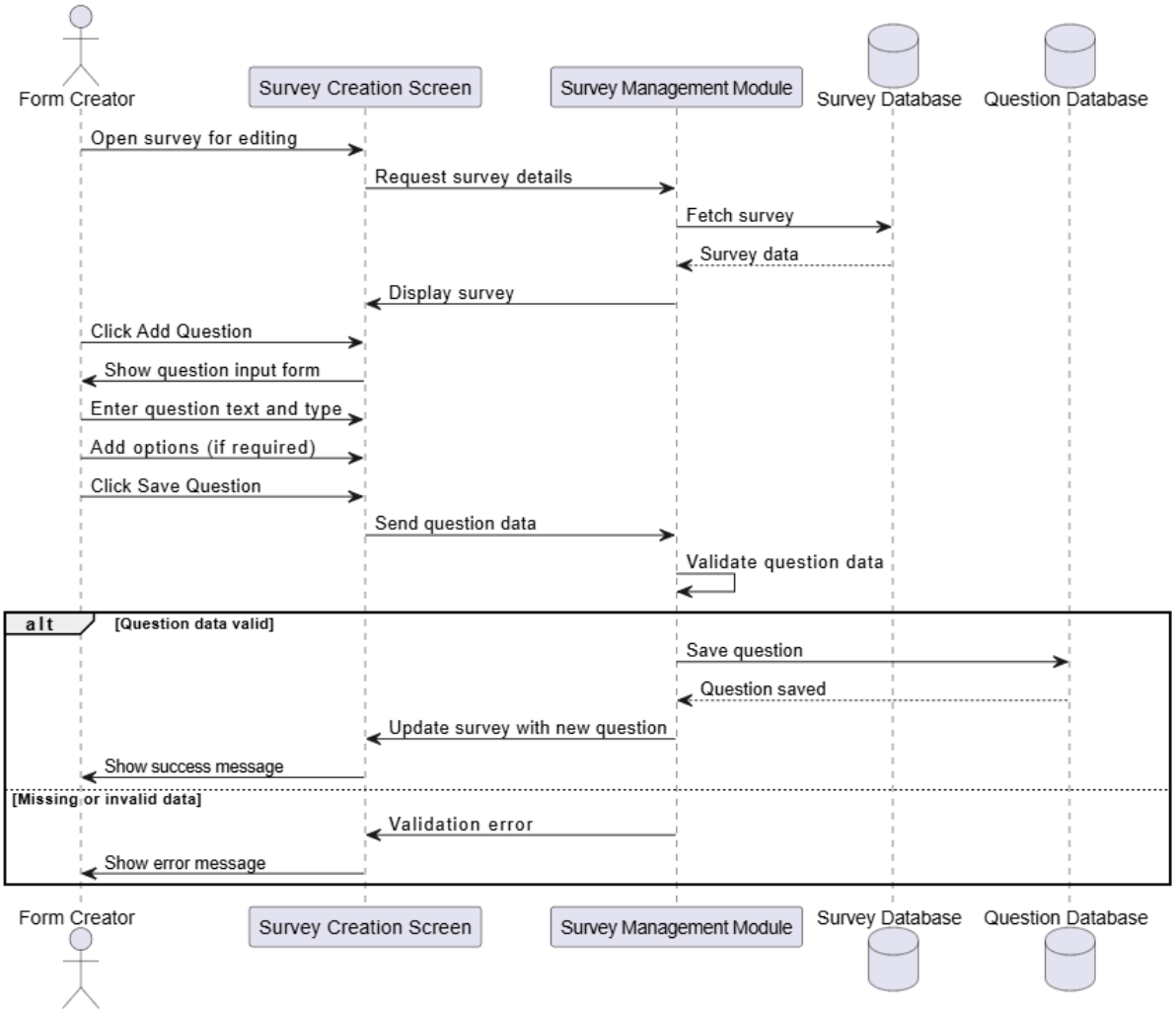


Figure 9 - Sequence Diagram to Add Questions

3.9 Edit/Update Survey (Form Creator Panel)

This sequence diagram explains how a form creator edits or updates an existing survey in the Access Form system. First, the form creator opens the “View Surveys” option and selects a survey to edit. The system retrieves the selected survey details from the database and displays them on the survey editing screen. The form creator updates the survey title, questions, or options as needed. After making changes, the form creator saves the updated survey or republishes it. The system validates the updated information and updates the survey record in the database. If the update is successful, the system shows a confirmation message. If any error occurs or required information is missing, the system shows an error message and asks the form creator to correct the data. This process allows surveys to be modified easily.

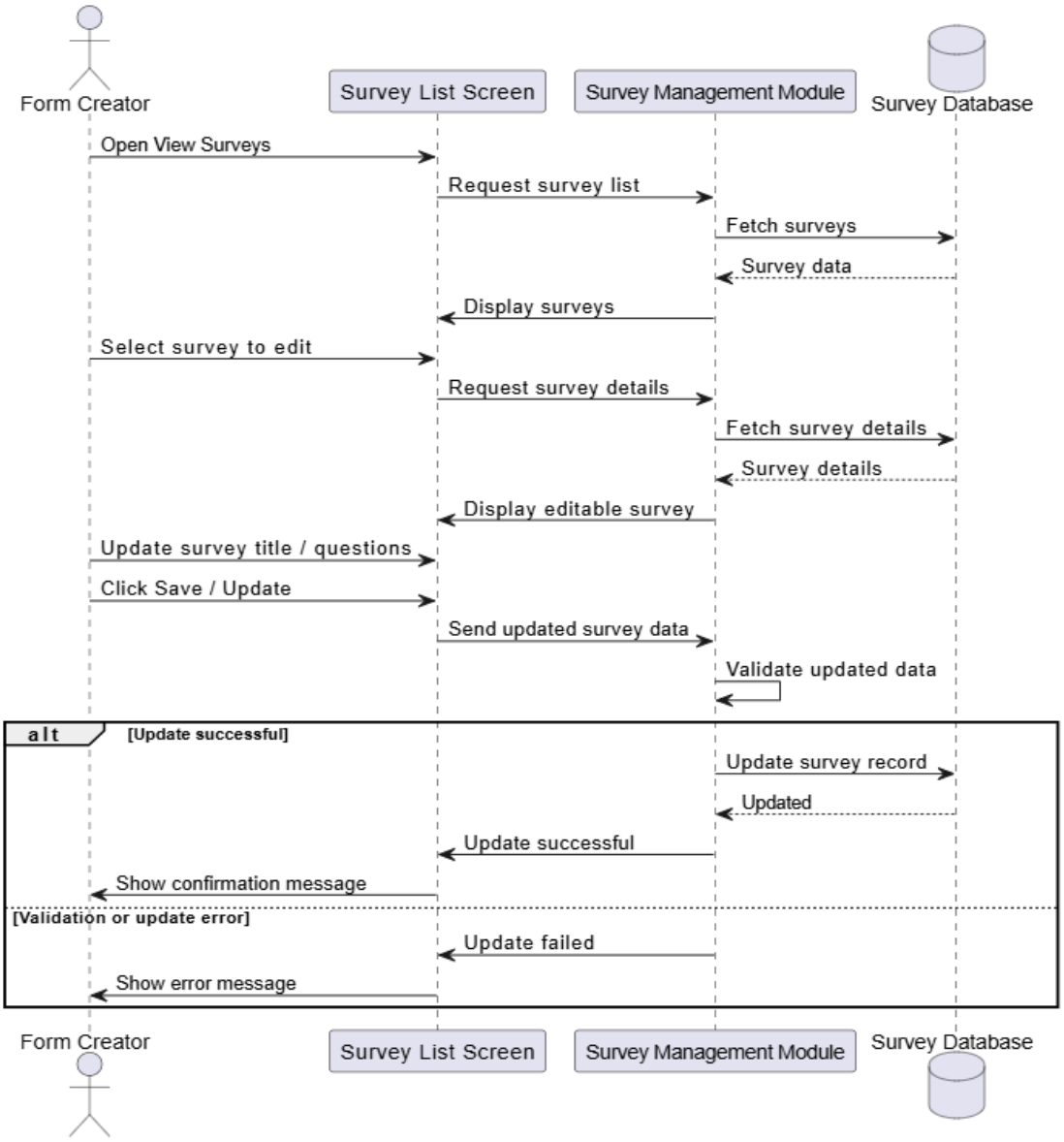


Figure 10 - Sequence Diagram to Edit / Update Survey

3.10 Delete Survey (Form Creator Panel)

This sequence diagram explains how a form creator deletes a survey in the Access Form system. First, the form creator opens the “View Surveys” option and selects the survey that needs to be deleted. The system asks for confirmation before deleting the survey to avoid accidental deletion. After confirmation, the system sends the delete request to the survey management process. The survey management process removes the survey record from the database. If the survey is deleted successfully, the system shows a confirmation message. If any error occurs during deletion, the system shows an error message. This process helps form creators safely remove surveys that are no longer required.

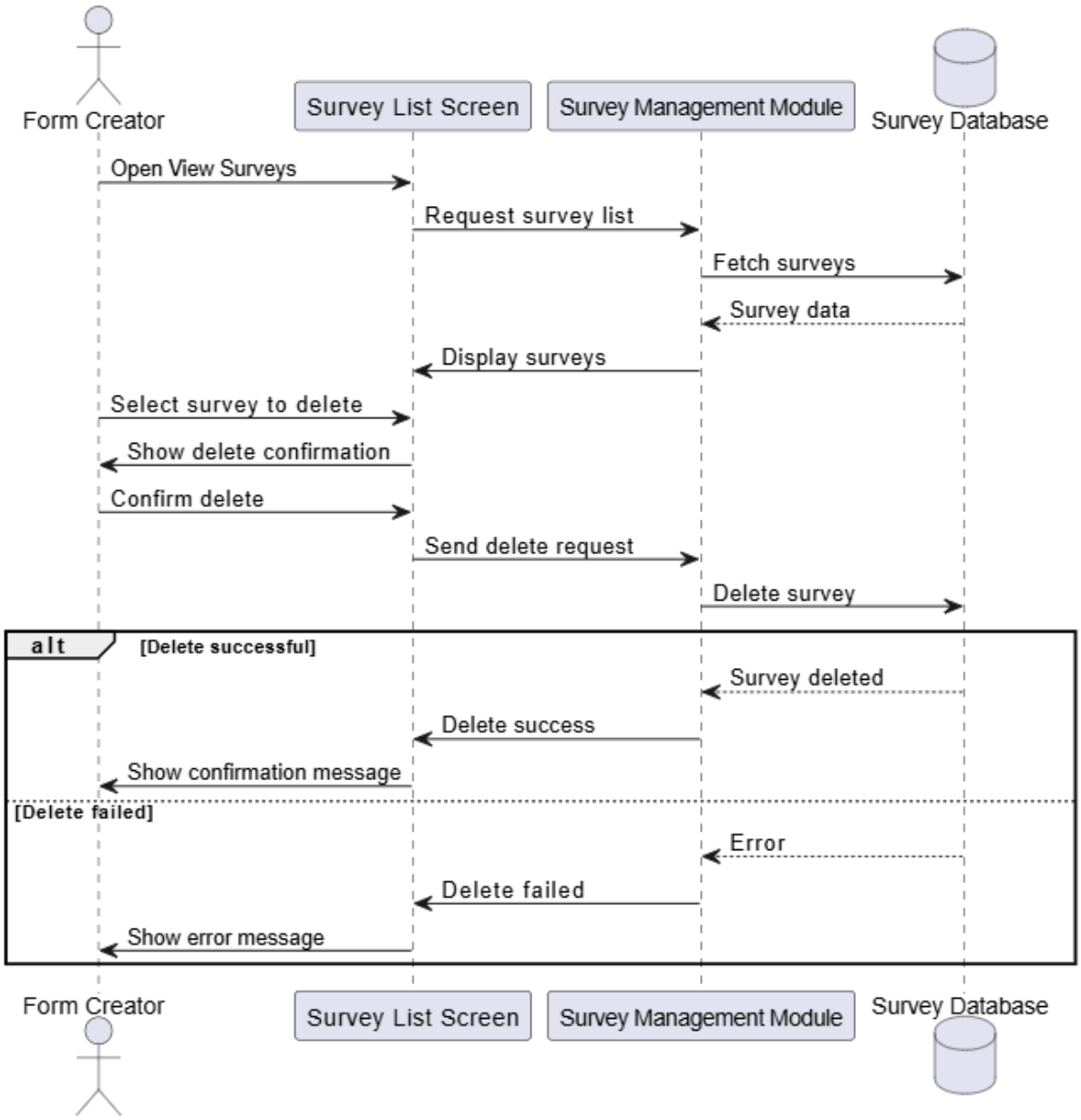


Figure 11 - Sequence Diagram to Delete Survey

3.11 Fill / Submit Survey (Respondent Panel)

This sequence diagram explains how a respondent fills and submits a survey in the Access Form system. First, the respondent opens a survey link or selects a survey from the available list. The system retrieves the survey questions from the database and displays them on the survey response screen. The respondent reads the questions and enters answers in the given fields. After completing the survey, the respondent clicks the submit button. The system validates the answers and stores the responses in the database. If the submission is successful, the system shows a thank-you or confirmation message. If any required question is left unanswered or an error occurs, the system shows an error message and asks the respondent to correct the response. This process allows respondents, including users with accessibility needs, to submit surveys easily.

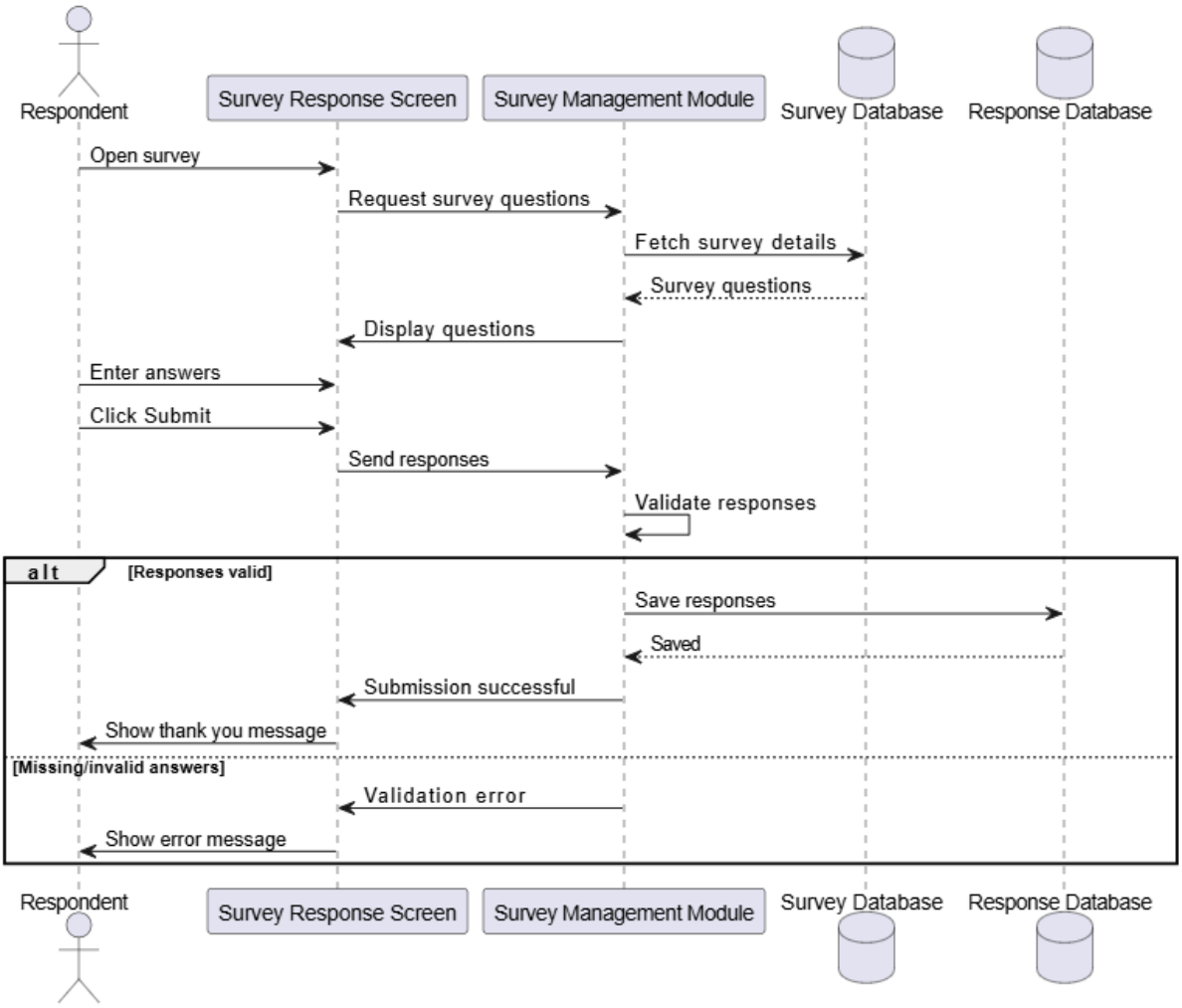


Figure 12 - Sequence Diagram to fill / Submit Survey

3.12 Apply Accessibility Settings

This sequence diagram explains how a user adjusts accessibility settings in the Access Form system. First, the user opens the “Accessibility Settings” option from the system interface. The system displays different accessibility options such as text size, high contrast mode, keyboard navigation, and screen reader support. The user selects or changes the required accessibility settings according to their needs. After applying the changes, the system saves the selected settings in the database. The interface is updated immediately based on the new accessibility preferences. If the settings are saved successfully, the system shows a confirmation message. If any error occurs while saving the settings, the system shows an error message. This process ensures that disabled users can use the system comfortably and independently.

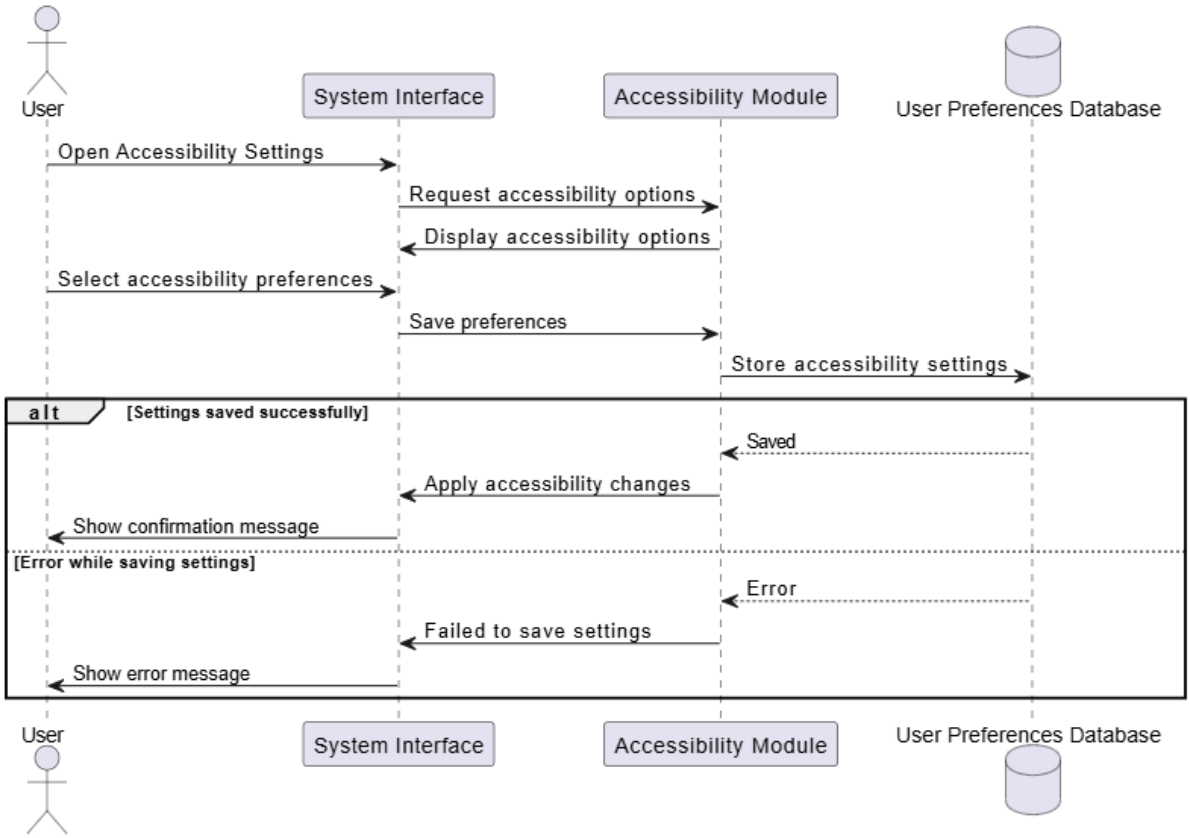


Figure 13 - Sequence Diagram to Apply Accessibility Settings

3.13 User Logout

This sequence diagram explains how a user logs out from the Access Form system. First, the logged in user clicks the logout option from the system interface. The system sends the logout request to the user management process. The system clears the active user session and ends the login state. After successful logout, the system redirects the user back to the login screen and shows a logout confirmation message. If any error occurs during the logout process, the system shows an error message to the user. This process ensures secure session termination for all users.

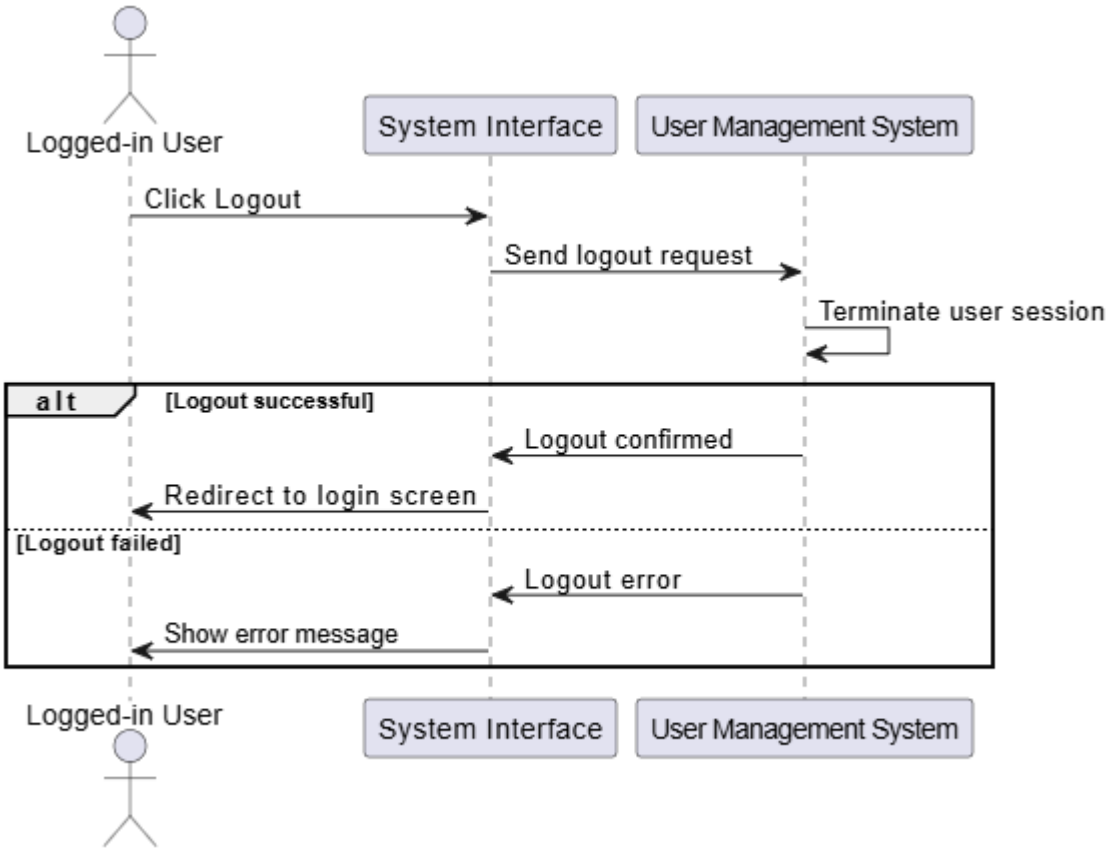


Figure 14 - Sequence Diagram for User Logout

4 Architecture Design Diagram

4.1 Three Tier Architecture

The Access Form system follows a three-tier architecture to make the system simple, organized, and easy to use for all users, especially disabled persons. The first layer is the Presentation Layer, which includes all user interfaces such as login and registration screens, dashboards, survey creation screens, survey response screens, and accessibility settings. This layer is responsible for user interaction and provides accessibility features like large text, high contrast mode, keyboard navigation, and screen reader support so that disabled users can easily use the system.

The second layer is the Application Layer, also called the Business Logic Layer. This layer handles all system operations such as user authentication, role management (admin, form creator, and respondent), survey creation and management, question handling, response processing, and applying accessibility rules. It validates user input, manages system workflows, and ensures that accessibility settings selected by users are properly applied throughout the system.

The third layer is the Data Layer, which is responsible for storing and managing data. This layer contains databases for users, surveys, questions, responses, and accessibility preferences. All data is securely stored and retrieved from this layer whenever required. Accessibility preferences of disabled users are also saved so that the system automatically adapts to their needs in future use. This three-tier architecture improves system performance, maintainability, and accessibility while keeping each layer independent and well-organized.

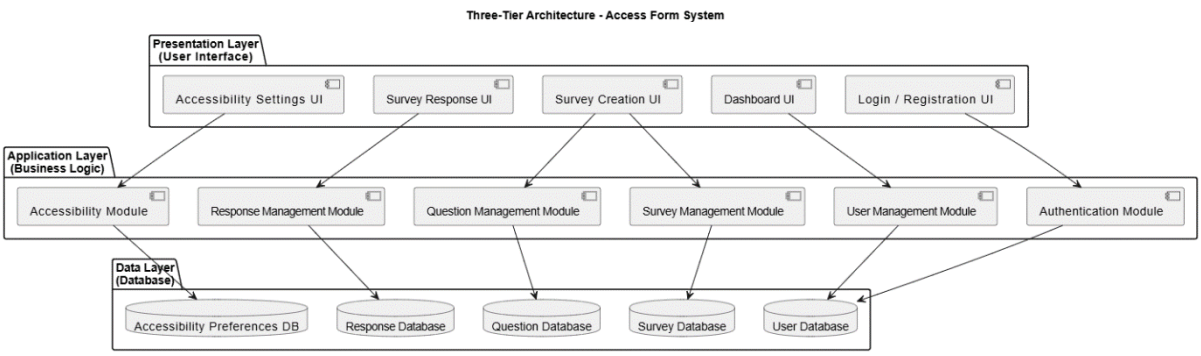


Figure 15 - Three Tier Architecture Diagram

4.2 System Architecture

The system architecture diagram illustrates the overall structure of the Access Form system. Users interact with the system through a web browser. The frontend handles user interfaces and accessibility features, while the backend server processes business logic and manages communication with the database. The database stores user information, surveys, and responses.

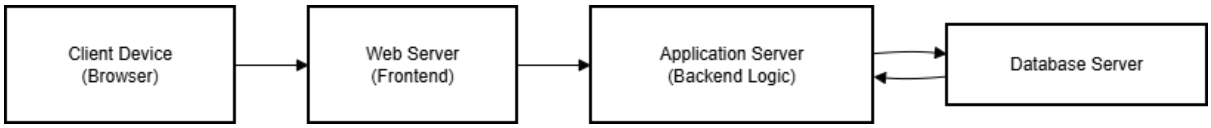
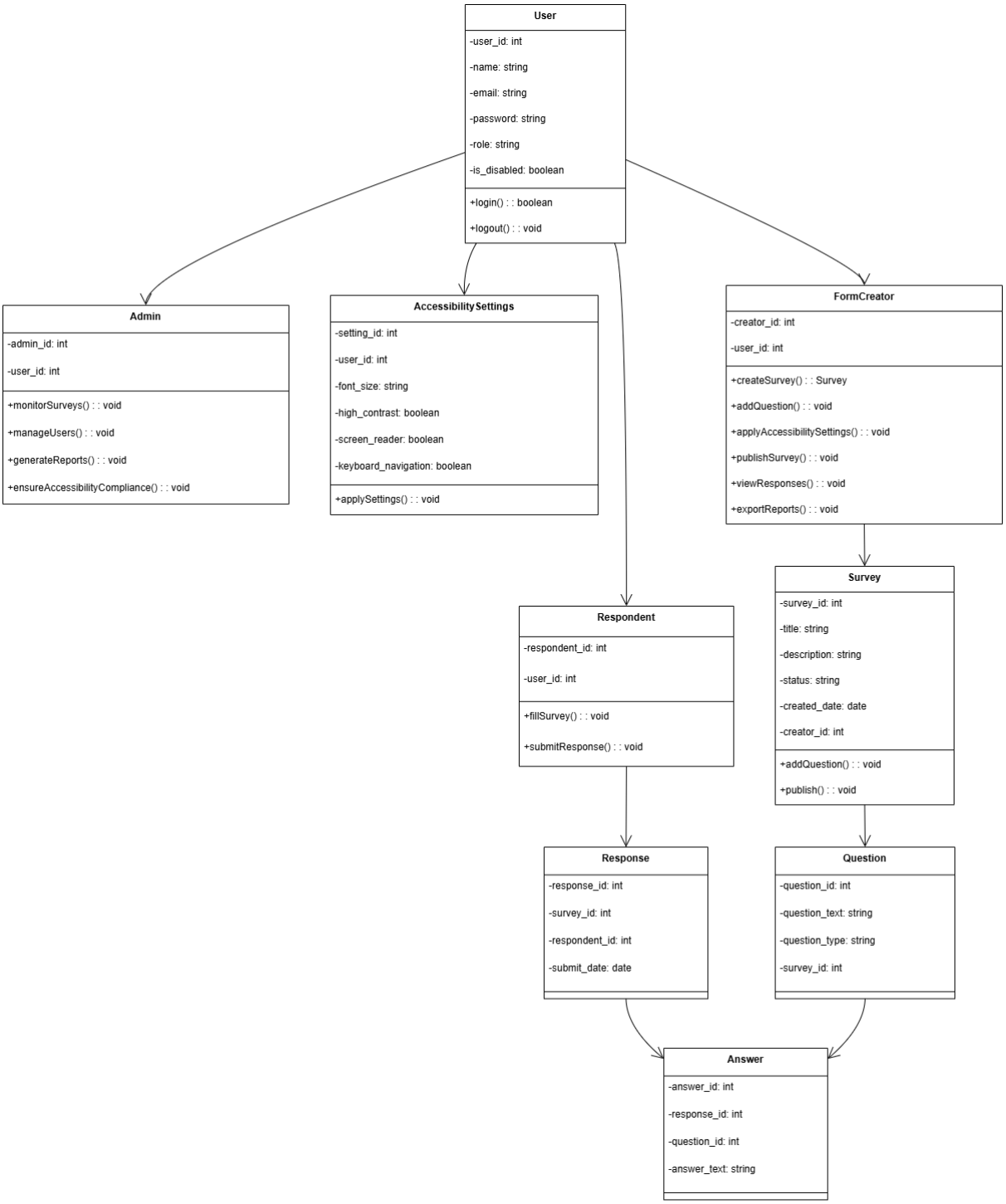


Figure 16 - System Architecture Diagram

5 Class Diagram

This class diagram shows how the **Access Form system** works in a simple way. There are three main user roles: **Admin**, **Form Creator**, and **Respondent**, all connected to the main **User** class. Admin manages users and surveys, Form Creator makes and publishes surveys with questions, and Respondent fills and submits surveys. The **Accessibility Settings** class is linked to User so that every person can adjust things like font size or screen reader support. A **Survey** contains many **Questions**, each question can get **Responses**, and every response has one or more **Answers**. In short, the diagram explains how users, surveys, questions, and answers are connected to make the system work smoothly.



6 Database Design

This database design explains how the survey system organizes its data in a clear way. The **User table** stores basic details like name, email, password, role, and status. From this table, three roles are linked: **Admin**, **Form Creator**, and **Respondent**, each with their own responsibilities. Admin manages users and surveys, Form Creator creates and publishes surveys, while Respondent fills and submits surveys. The **Accessibility Settings table** is connected to User so that special features like screen reader, high contrast, and keyboard navigation can be applied. A **Survey table** holds survey details and is linked to multiple **Questions**. Respondents submit their answers, which are stored in the **Response table**, and each response is further detailed in the **Answer table**. Together, these tables show how users, surveys, questions, and answers are connected to make the system work smoothly and support accessibility for all.

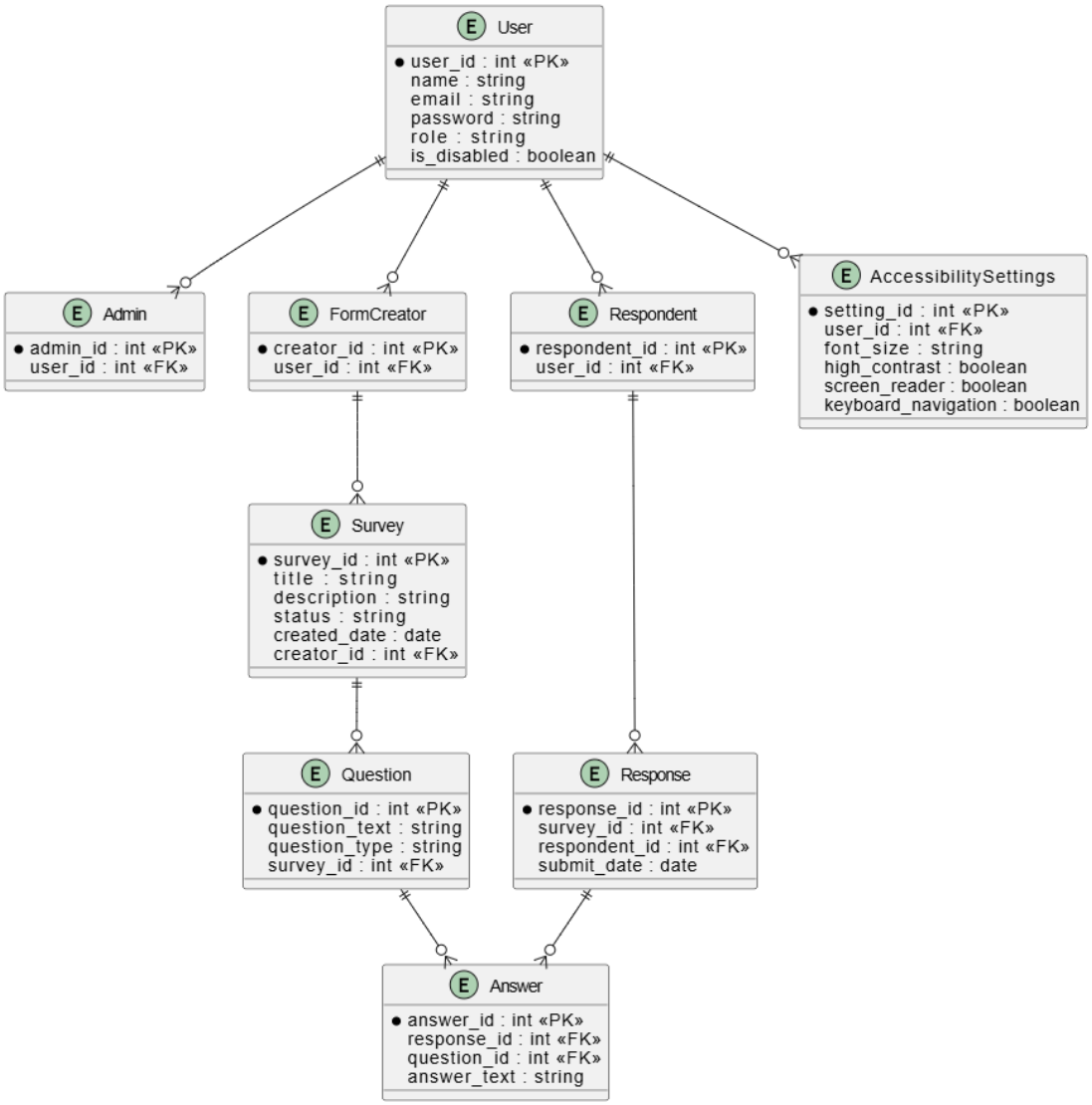


Figure 17 - Database Design Diagram

7 Interface Design

7.1 Login Registration

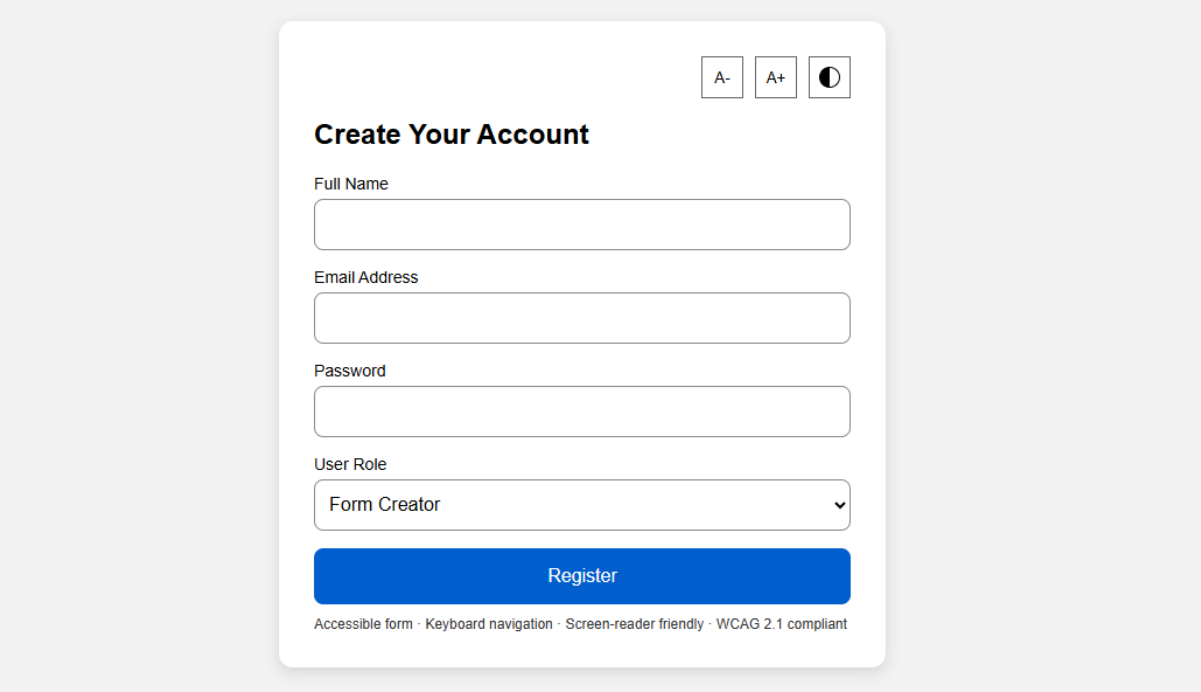


Figure 18 - Interface Diagram for User Registration

7.2 User Login

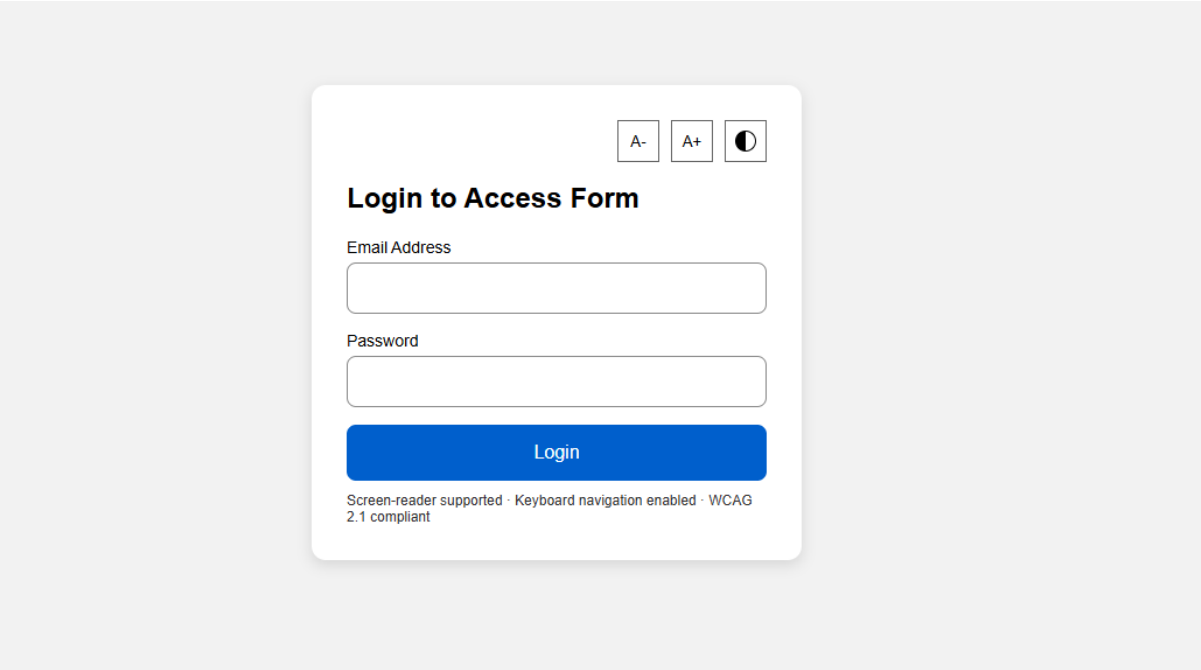


Figure 19 - Interface Design for User Login

7.3 ***Dashboard***

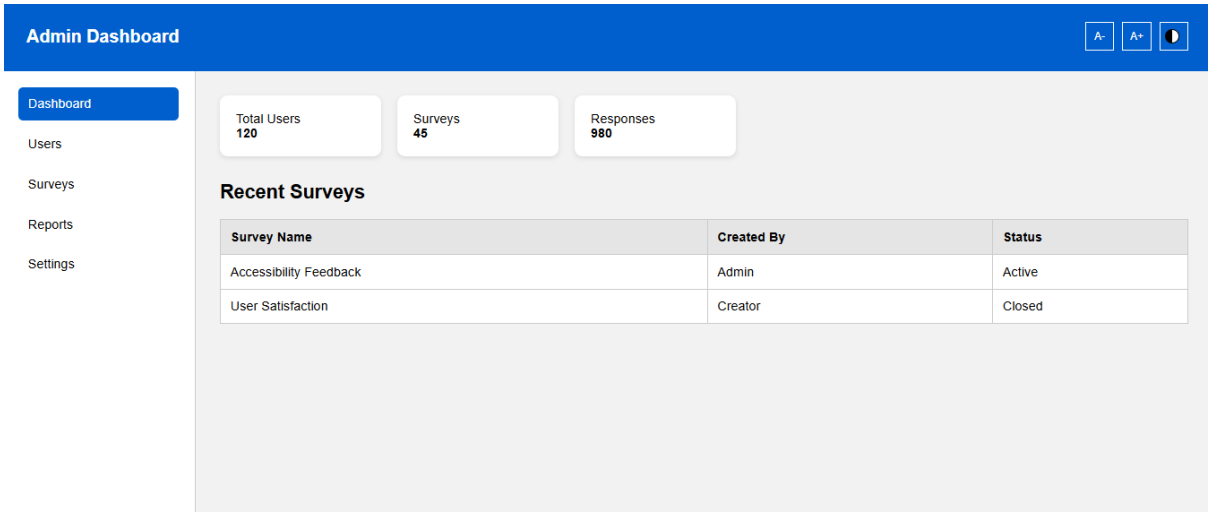


Figure 20- Interface Design for Dashboard

7.4 ***Survey Creation***

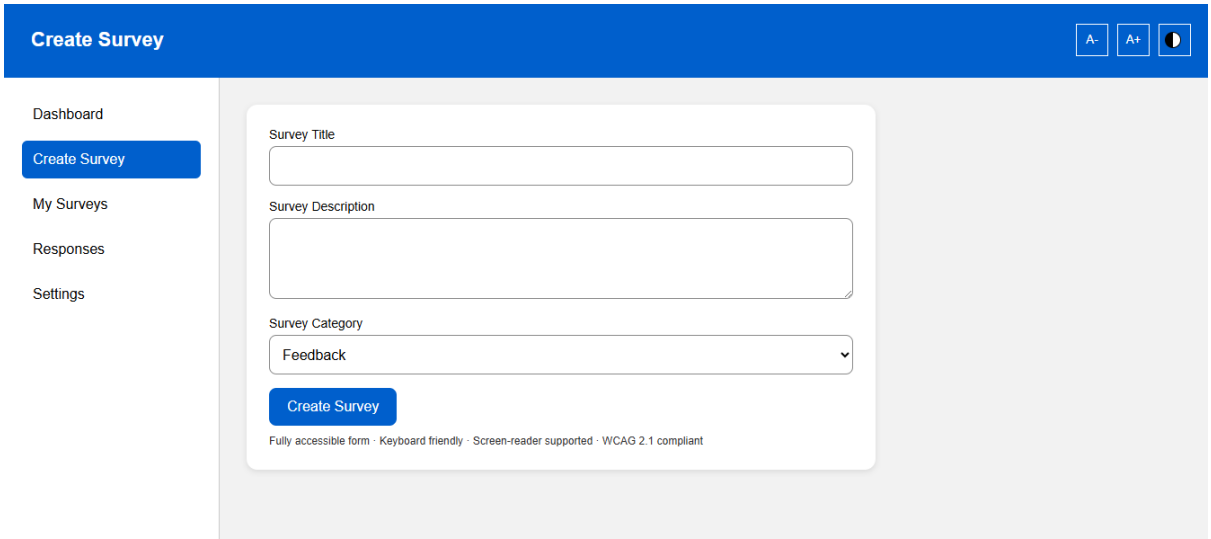


Figure 21 - Interface Design for Survey Creation

7.5 ***Add Question***

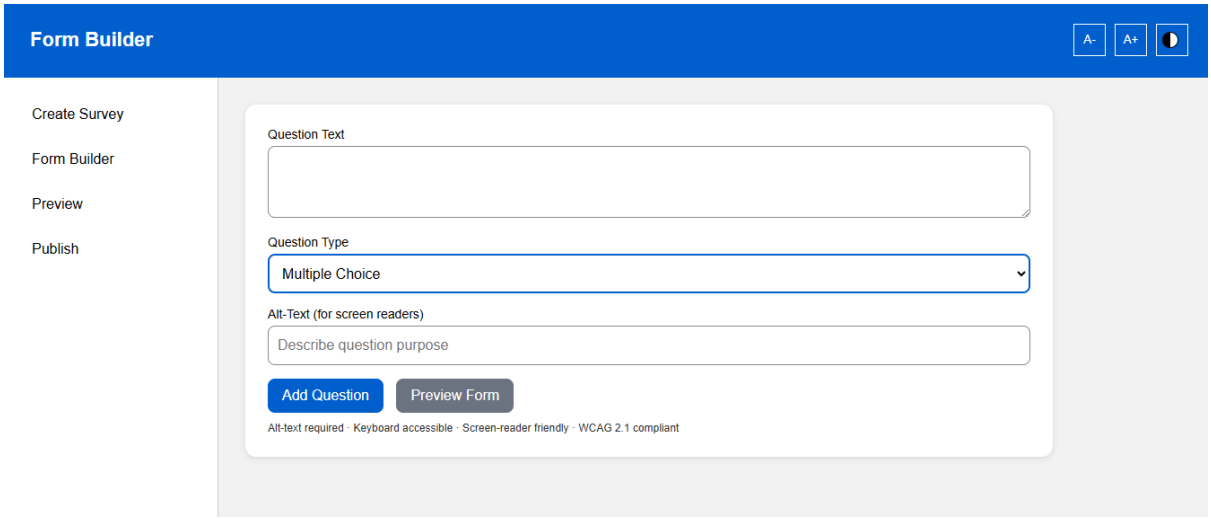


Figure 22 - Interface Design to Add Question

7.6 ***View Survey Screen***

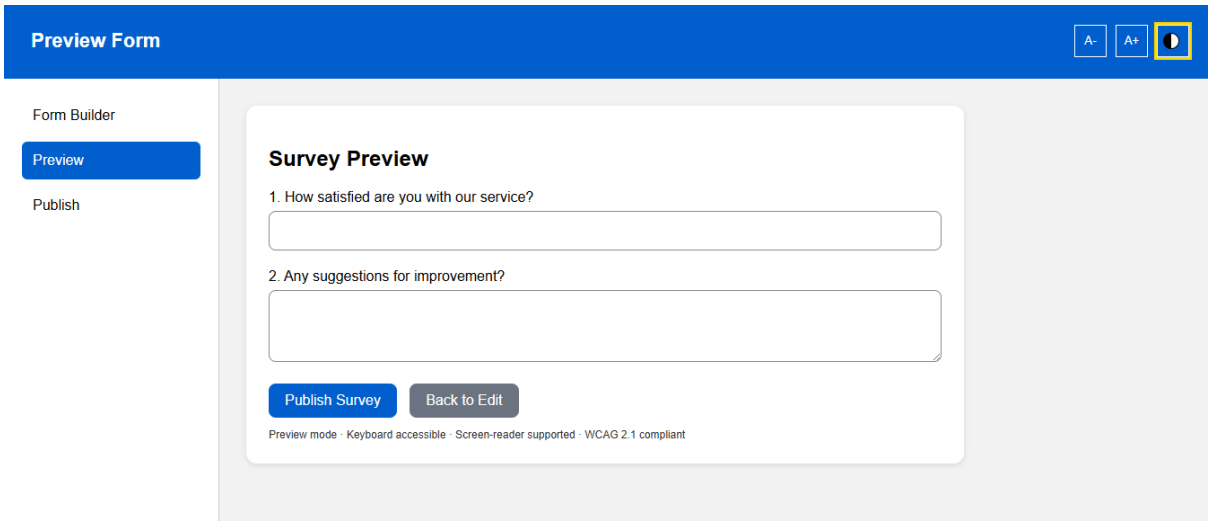


Figure 23 - Interface Design to view Survey Screen

7.7 ***Start Survey Screen***

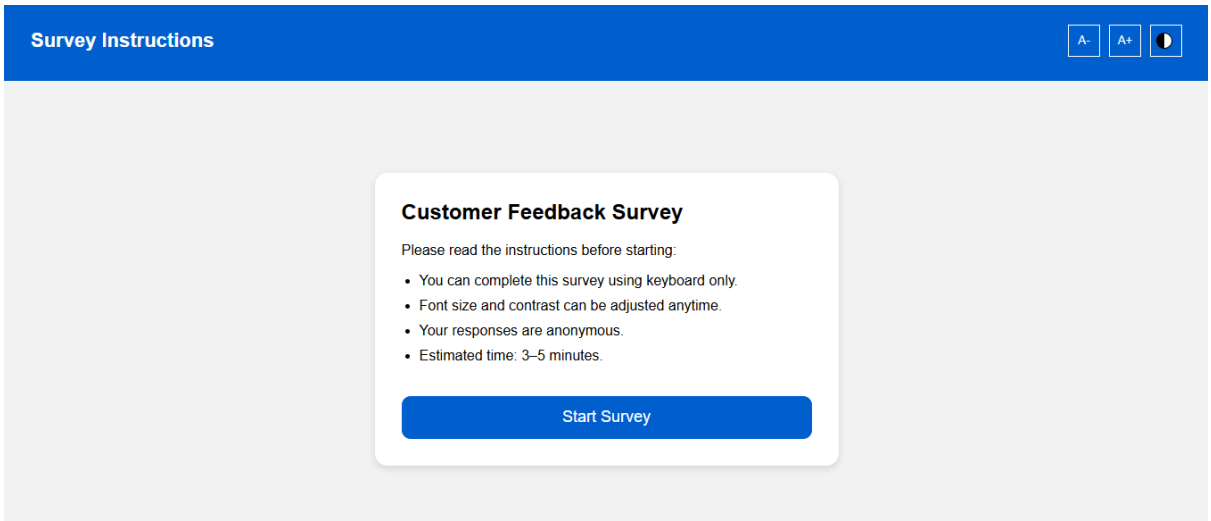


Figure 24 - Interface Design to Start Survey

7.8 ***Survey Questions Screen***

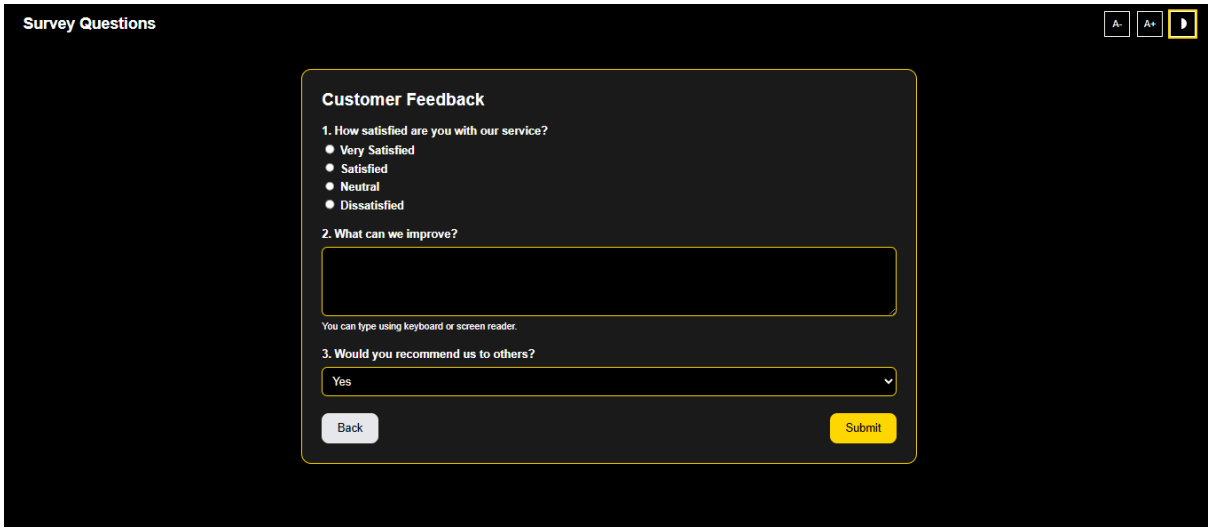


Figure 25 - Interface Diagram to Fill Survey Questions

7.9 *Survey Submission Screen*

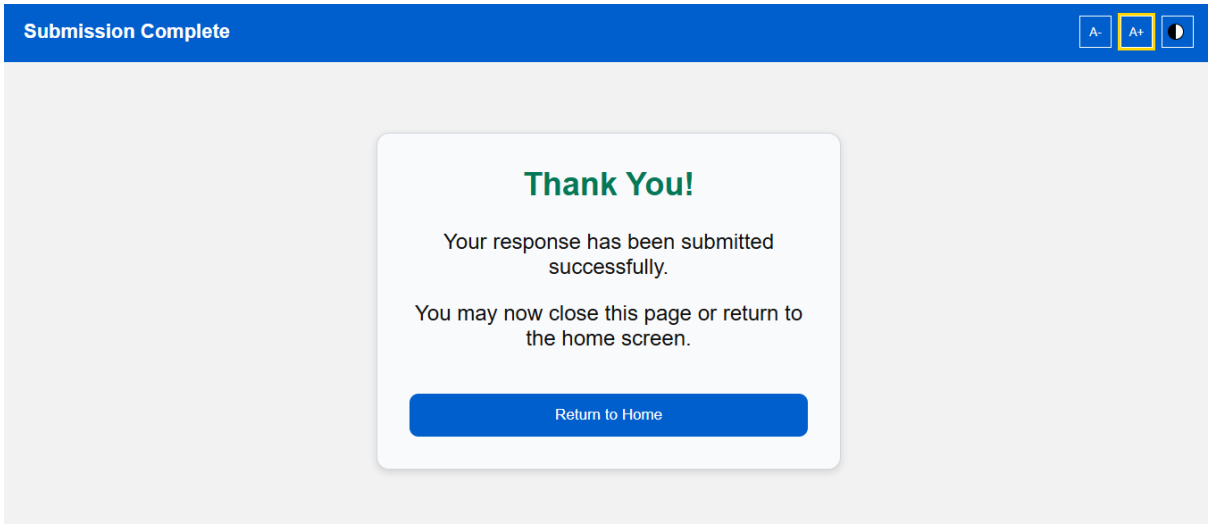


Figure 26 - Interface Design For Survey Submission

7.10 *View Surveys*

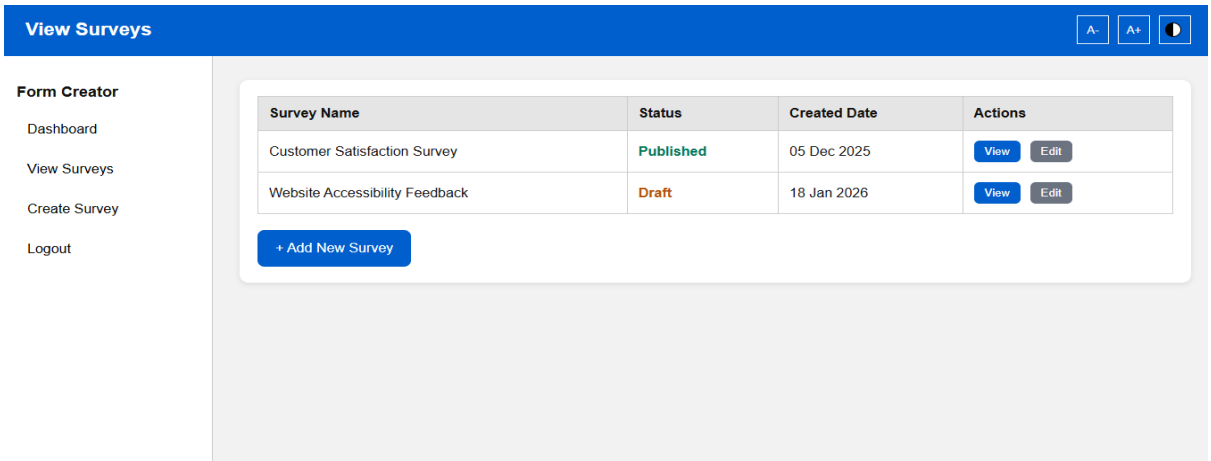


Figure 27 - Interface Design to View Surveys

7.11 Export Survey Screen

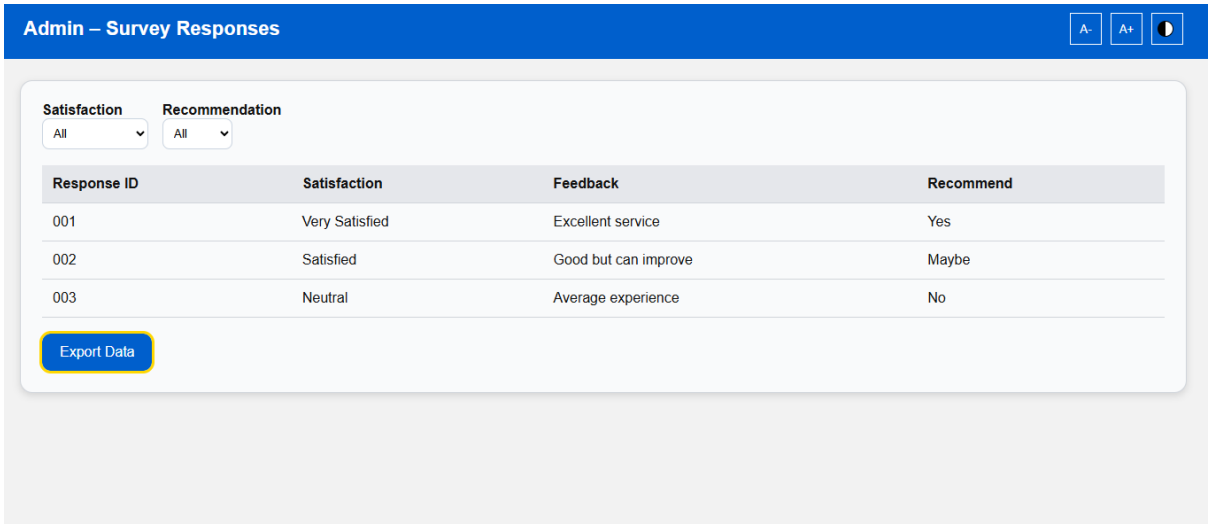


Figure 28 - Interface Design to Export Report

8 Test Cases

8.1 User Login

Table 1 - Valid Table

Field	Details
Test Case ID	TC-01
Use Case ID	UC-AP-01 / UC-FC-01 / RUC-01
Description	Verify that a registered user can log in successfully.
Preconditions	User account must exist
Test Steps	• Open login page • Enter Valid Credentials • Click Login
Expected Result	User is redirected to the dashboard
Status	Pass

Table 2 - Exception Table

Field	Details
Use Case ID	UC-AP-01 / UC-FC-01 / RUC-01
Precondition	User is on Login screen
Test Steps	• Enter wrong email/password • Click Login
Expected Result	• Error message shown • Login denied
Exception Type	Authentication Error

8.2 Create New Survey

Table 3 - Valid Table

Field	Details
Test Case ID	TC-02
Use Case ID	UC-FC-01
Description	Verify Form Creator can create a new survey
Preconditions	User logged in as Form Creator
Test Steps	• Click “Create Survey” • Enter survey title • Save
Expected Result	New survey is created successfully
Status	Pass

Table 4 - Exception Table

Field	Details
Use Case ID	UC-FC-01
Precondition	Form Creator Logged in
Test Steps	• Leave survey title empty • Click Save
Expected Result	Validation message “Title is required”
Exception Type	Input Validation Error

8.3 Add Questions to Survey

Table 5 - Valid Table

Field	Details
Test Case ID	TC-03
Use Case ID	UC-FC-02
Description	Verify questions can be added to a survey
Preconditions	Survey already exists
Test Steps	• Open survey • Click “Add Question” • Enter Question Details
Expected Result	Question is added Successfully
Status	Pass

Table 6 - Exception Table

Field	Details
Use Case ID	UC-FC-02
Precondition	Form Creator Logged in, Survey Opened
Test Steps	• Leave question text empty • Click “Add Question”
Expected Result	Validation Error “Question text is required”
Exception Type	Input Validation Error

8.4 Apply Accessibility Settings

Table 7 - Valid Table

Field	Details
Test Case ID	TC-04
Use Case ID	UC-FC-03
Description	Verify accessibility settings can be applied
Preconditions	Survey exists
Test Steps	• Open accessibility settings • Enables Options • Save
Expected Result	Accessibility Settings Applied
Status	Pass

Table 8 - Exception Table

Field	Details
Use Case ID	UC-FC-03
Precondition	User using keyboard only
Test Steps	• Navigating form using Tab key
Expected Result	All elements reachable via keyboard
Exception Type	Accessibility Failure (WCAG 2.1)

8.5 Publish Survey

Table 9 - Valid Table

Field	Details
Test Case ID	TC-05
Use Case ID	UC-FC-04
Description	Verify Survey can be published
Preconditions	Survey is complete
Test Steps	• Review survey • Click “Publish”
Expected Result	Survey is published and availble to respondents
Status	Pass

Table 10 - Exception Table

Field	Details
Use Case ID	UC-FC-04
Precondition	Survey created
Test Steps	• Click Publish without adding question
Expected Result	System prevents publish, warning shows
Exception Type	Business Rules Violation

8.6 View Survey Responses

Table 31 - Valid Table

Field	Details
Test Case ID	TC-06
Use Case ID	UC-FC-05
Description	Verify Form Creator can view responses
Preconditions	Responses must exist
Test Steps	• Open “View Responses”
Expected Result	Responses are displayed
Status	Pass

8.7 Export Survey Reports

Table 42 - Valid Table

Field	Details
Test Case ID	TC-07
Use Case ID	UC-FC-06
Description	Verify reports can be exported
Preconditions	Survey data availble
Test Steps	• Select export option • Click “Export”
Expected Result	Report file is downloaded
Status	Pass

8.8 Monitor Surveys (Admin Panel)

Table 53 - Valid Table

Field	Details
Test Case ID	TC-08
Use Case ID	UC-AP-01
Description	Verify admin can monitor surveys
Preconditions	Admin logged in
Test Steps	• Open survey monitoring panel
Expected Result	List of surveys displayed
Status	Pass

8.9 Manage Users (Admin Panel)

Table 64 - Valid Table

Field	Details
Test Case ID	TC-09
Use Case ID	UC-AP-02
Description	Verify admin can manage users
Preconditions	Admin logged in
Test Steps	• Add/Edit/Delete User
Expected Result	User information updated
Status	Pass

8.10 Generate Reports (Admin Panel)

Table 75 - Valid Table

Field	Details
Test Case ID	TC-10
Use Case ID	UC-AP-03
Description	Verify admin can generate reports
Preconditions	Survey data available
Test Steps	• Select report option
Expected Result	Reports generated successfully
Status	Pass

Table 16 - Exception Table

Field	Details
Use Case ID	UC-AP-03
Precondition	No responses available
Test Steps	• Click Export
Expected Result	Error Message “No data to export”
Exception Type	Empty Dataset Handling

8.11 Accessibility Compliance Check

Table 87 - Valid Table

Field	Details
Test Case ID	TC-11
Use Case ID	UC-AP-04
Description	Verify accessibility compliance checking
Preconditions	Survey exists
Test Steps	• Run compliance check
Expected Result	Accessibility issues identified
Status	Pass

8.12 Fill Survey (Respondent Panel)

Table 98 - Valid Table

Field	Details
Test Case ID	TC-12
Use Case ID	RUC-01
Description	Verify respondent can fill survey
Preconditions	Survey is published
Test Steps	• Open survey • Fill answer
Expected Result	Survey filled successfully
Status	Pass

Table 19 - Exception Table

Field	Details
Use Case ID	RUC-01
Precondition	Screen reader enabled
Test Steps	• Navigate Questions
Expected Result	Labels read correctly
Exception Type	Assistive Technology Failure

8.13 Submit Survey Response

Table 20 - Valid Table

Field	Details
Test Case ID	TC-12
Use Case ID	RUC-02
Description	Verify survey submission
Preconditions	Survey filled
Test Steps	• Click “Submit”
Expected Result	Response submitted successfully
Status	Pass

Table 101 - Exception Table

Field	Details
Use Case ID	RUC -01
Precondition	Survey opened
Test Steps	• Skip required questions • Click Submit
Expected Result	• Error message shown • Submission blocked
Exception Type	